

IT-24655

Q: Prove that the set of rational numbers $\mathbb{Q}$, equipped with the two binary operations of addition & multiplication, form a field.

→ We take the rational numbers $\mathbb{Q}$ to be the set of equivalence classes of ordered pairs (a, b) with $a, b \in \mathbb{Z}$ and $b \neq 0$, where $(a, b) \sim (a', b')$ if $ab' = a'b$. We identify the class of (a, b) with the usual fraction $\frac{a}{b}$. Define addition & multiplication in the usual way:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

for $b \neq 0, d \neq 0$, below we show these operations make $\mathbb{Q}$ a field

1. We can check that if $\frac{a}{b} = \frac{a'}{b'}$ & $\frac{c}{d} = \frac{c'}{d'}$ then

$$\frac{ad+bc}{bd} = \frac{a'd'+b'c'}{b'd'} \quad \& \quad \frac{ac}{bd} = \frac{a'c'}{b'd'}$$

From $\frac{a}{b} = \frac{a'}{b'}$ & $\frac{c}{d} = \frac{c'}{d'}$ we have $ab' = a'b$ & $cd = c'd'$

Compute,

$$(ad+bc)b'd' = (ab')(d'd') + (bc)(b'd') = (a'b)(d'd') + bc(b'd')$$

2. $(\mathbb{Q}, +)$ is an abelian group.

Take any $\frac{a}{b}, \frac{c}{d} \neq \frac{e}{f} \in \mathbb{Q}$.

Closure : $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ is a rational number

since bd

$$\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{ad+bc}{bd} + \frac{e}{f} = \frac{f(ad+bc) + e(bd)}{(bd)f}$$

similar expansion for $\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right)$: both give the same numerator by associativity of integer operation

Identity: $0 = \frac{0}{1}$ satisfy $\frac{a}{b} + 0 = \frac{a}{b}$

Inverse: additive inverse of $\frac{a}{b}$ is $-\frac{a}{b} = -\frac{a}{b}$

because, $\frac{a}{b} + -\frac{a}{b} = \frac{0}{b} = 0$

Commutativity: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd} = \frac{bc+ad}{db} = \frac{c}{d} + \frac{a}{b}$

Thus, $(\mathbb{Q}, +)$ is an abelian group

3. Multiplication on $\mathbb{Q} \setminus \{0\}$ is an abelian group

Closure: product $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ is

rational since $bd \neq 0$

Associativity & commutativity: follow from associativity & commutativity of integer multiplication.

$$\begin{aligned} \left(\frac{a}{b} \cdot \frac{c}{d}\right) \cdot \frac{e}{f} &= \frac{ac}{bd} \cdot \frac{e}{f} = \frac{(ac)e}{bd \cdot f} = \frac{ac \cdot e}{df \cdot b} = \frac{a}{b} \cdot \frac{ce}{df} \\ &= \frac{a}{b} \left(\frac{c}{d} \cdot \frac{e}{f}\right) \end{aligned}$$

Multiplicative Identity : $1 = \frac{1}{1}$ satisfies $\frac{a}{b} \cdot 1 = \frac{a}{b}$

Distributivity: For addition & multiplication

$$\begin{aligned}\frac{a}{b} \cdot \left(\frac{c}{d} + \frac{e}{f} \right) &= \frac{a}{b} \cdot \frac{cf+ed}{df} = \frac{a(cf+ed)}{b \cdot df} = \frac{acf+aed}{bdf} \\ &= \frac{ac}{bd} + \frac{ae}{bf} = \frac{a}{b} \cdot \frac{c}{d} + \frac{a}{b} \cdot \frac{e}{f}\end{aligned}$$

4. Multiplicative inverse exist for non zero rationals. Take a non zero rational $\frac{a}{b}$, Its multiplicative inverse is $\frac{b}{a}$ because.

$$\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ba} = \frac{1}{1} = 1$$

We must check their inverse is well defined: